Vapnik-Chervonenkis Dimension and Probabilistic Bounds

MATH 598 Presentation

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Classifiers and Misclassification

Let $\mathcal{D}_n := \{(\boldsymbol{x}_1, y_1), \cdots, (\boldsymbol{x}_n, y_n)\}$ be the training set, $(\boldsymbol{x}_i, y_i)$ are jointly distributed random variables from the probability space $\mathcal{X} \times \mathcal{Y}$. We consider a binary classification problem where $\mathcal{Y} = \{0, 1\}$. A **classifier** is a Borel-measurable function $f : \mathcal{X} \to \mathcal{Y}$. Our goal is to find the "best" classifier based on the data we have and make predictions on other data. Misclassification occurs when $f(\boldsymbol{x}_i) \neq y_i$, and we may ask the following:

- How likely will a classifier fits the training sample well, but performs poorly outside the traing sample?
- How large will the training sample be in order to have a low misclassification rate overall?
- In general how can we control the misclassification rate under a certain level?

To answer those questions, we will study the theory of Vapnik-Chervonenkis (VC) dimension and the probabilistic bounds derived from it.

Constructing Binary Classifiers

Let $f : \mathcal{X} \to \{0, 1\}$ be a binary classifier, let $S \subset \mathcal{X}$, we define

$$f\Big|_S : S \to \{0,1\}, \quad f\Big|_S(\boldsymbol{x}) = f(\boldsymbol{x}), \forall \boldsymbol{x} \in S$$

as the restriction of f to S. Let $\mathcal{C}$ be the class of all classifiers, we define

$$\Pi_{\mathcal{C}}(S) := \left\{ f\Big|_S : f \in \mathcal{C} \right\}$$

as the number of distinct restrictions of classifiers in $\mathcal{C}$ defined by S, i.e, the set of all possible dichotomies on S:

$$\Pi_{\mathcal{C}}(S) := \{(f(\boldsymbol{x}_1), \cdots, f(\boldsymbol{x}_m)), f \in \mathcal{C})\}$$

where we assume $S := \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_m\}$ and $f(\boldsymbol{x}_i) \in \{0, 1\}$. Clearly $|\Pi_{\mathcal{C}}(S)| \leq 2^{|S|}$.

Vapnik-Chervonenkis Dimension

Definition: A finite set $S \subset \mathcal{X}$ is *shattered* by $\mathcal{C}$, if $|\Pi_{\mathcal{C}}(S)| = 2^{|S|}$. The *Vapnik-Chervonenkis Dimension* (VCD) of $\mathcal{C}$, denote as $\mathbf{VCD}(\mathcal{C})$ (or simply V) is the cardinality d of the largest set S shattered by $\mathcal{C}$. We say $\mathbf{VCD}(\mathcal{C}) = \infty$ if $\mathcal{C}$ shatters arbitrarily sets.

- In other words, $S = \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_m\} \subset \mathcal{X}$ is shattered by $\mathcal{C}$ if for every assignment of labels to those points, there exists $f \in \mathcal{C}$ such that it makes no error when evaluating that set of data points.

We also define the **Growth Function** as

$$\Pi_C(m) = \max\{|\Pi_{\mathcal{C}}(S)|, |S| = m\}$$

where $m \in \mathbb{N}$. If $m \leq \mathbf{VCD}(\mathcal{C})$, we simply have $\Pi_{\mathcal{C}}(m) = 2^m$.

Some Examples:

- **Interval Classifiers:** Consider the class of classifiers $f \in \mathcal{C}$ as intervals in $\mathbb{R}$. We then claim that $\textbf{VCD}(\mathcal{C}) = 2$, i.e. an interval classifier can shatter any sets of two points.

Figure: Any interval can shatter 2 points

Figure: No interval can shatter 3 points of this pattern

Examples of Shattering and VC-Dimension

Some Examples:

- **Rectangles in $\mathbb{R}^2$:** Consider $\boldsymbol{x} \in \mathbb{R}^2$ and define $\mathcal{C}$ as the class of all rectangles:

$$[\alpha_1, \beta_1] \times [\alpha_2, \beta_2]$$

  We may assume any three points are not co-linear because if so then it reduces to the case of an interval classifier.

- We claim that $\textbf{VCD}(\mathcal{C}) = 4$.

Examples of Shattering and VC-Dimension

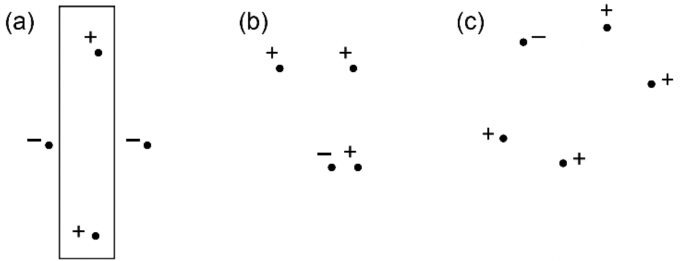

Figure: There exists a pattern of four points that can be shattered by rectangale, but not all patterns are possible. However no rectangle can shatter 5 points.

Examples of Shattering and VC-Dimension

Some Examples:

- **Linear Threshold Functions:** Consider $\boldsymbol{x} \in \mathbb{R}^n$ and define $\mathcal{C}$ as the class of all linear half spaces: $\boldsymbol{x}^\top \boldsymbol{\beta} > c$
- We claim that $\mathbf{VCD}(\mathcal{C}) = n + 1$.

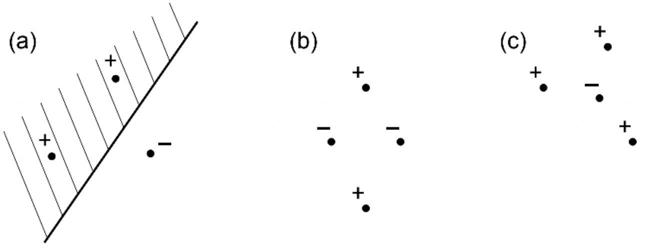

Figure: An example with $n = 2$. A linear half plane can shatter any 3 points but can not shatter any 4 points.

Examples of Shattering and VC-Dimension

Some Examples:

- **Convex d-gons in $\mathbb{R}^2$:** Consider $\mathcal{C}$ whose boundary is a convex d-gon in the plane.
- We claim that $\mathbf{VCD}(\mathcal{C}) = 2d + 1$.
- The idea is, we pick $2d + 1$ points on a circle, and for any labeling of those points, we can insert a convex polygon that either:
  - Has vertices as some of those points
  - Has edges tangent to the circle at some of those points

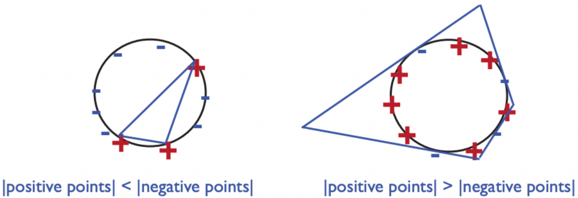

Figure: A 4-gon can shatter any 9 points on a circle

Properties of Growth Function

We have seen that the growth function satisfies $\Pi_{\mathcal{C}}(m) = 2^m$ if $m \leq \mathbf{VCD}(\mathcal{C})$ What if $m > \mathbf{VCD}(\mathcal{C})$? We define a function $\Phi_d(m)$ such that $\Phi_d(0) = 1, \Phi_0(m) = 1$ for all $d, m \in \mathbb{N}$ and

$$\Phi_d(m) = \Phi_d(m-1) + \Phi_{d-1}(m-1)$$

for all $d, m > 0$. Then we have

$$\Pi_{\mathcal{C}}(m) \leq \Phi_d(m)$$

if $\mathbf{VCD}(\mathcal{C}) = d$. The proof is done using induction on d, m. Also for all m, d,

$$\Phi_d(m) = \sum_{i=0}^{d} \binom{m}{i}$$

Properties of Growth Function

Finally, we can bound $\Pi_{\mathcal{C}}(m)$ by the following: When $m \geq d$, we have

Lemma (Sauer's Lemma)

Let $\mathbf{VCD}(\mathcal{C}) = d$*, for* $m \geq d$*, we have*

$$\Pi_{\mathcal{C}}(m) \leq \left(\frac{em}{d}\right)^d = \mathcal{O}(m^d).$$

Misclassification Rate

Suppose f is the optimal classifier, $f' \in \mathcal{C}$ be any classifier, let

$$(f' \oplus f)(\boldsymbol{x}) = \begin{cases} 1 & \text{if } f'(\boldsymbol{x}) \neq f(\boldsymbol{x}) \\ 0 & \text{otherwise} \end{cases}$$

- Define $err(f') = \mathbb{P}[(f \oplus f')(\boldsymbol{x}) = 1]$ as the *misclassification rate*, and $\Delta_f(\mathcal{C}) = \{f' \oplus f, f' \in \mathcal{C}\}$. We claim that $\mathbf{VCD}(\Delta_f(\mathcal{C})) = \mathbf{VCD}(\mathcal{C})$, and let
$$\Delta_{f,\epsilon}(\mathcal{C}) = \{\tilde{f} \in \Delta_f(\mathcal{C}), \mathbb{P}[\tilde{f}(\boldsymbol{x}) = 1] \geq \epsilon\}$$
i.e, the set of all classifiers in $\Delta_f(\mathcal{C})$ with misclassification rate at least ϵ.

ϵ-net

Definition: We say S is an ϵ-*net* for $\Delta_f(\mathcal{C})$ if $\forall \tilde{f} \in \Delta_{f,\epsilon}(\mathcal{C})$, $\exists \boldsymbol{x} \in S$ such that $\tilde{f}(\boldsymbol{x}) = 1$.

- It means that, if a classifier f' is "bad enough" that has misclassification rate of at least ϵ, then there must exist one point $\boldsymbol{x}$ such that f' misclassifies the label of $\boldsymbol{x}$;
- Conversely, if the training set S is an ϵ-net, then any classifier f' that fits the sample well (with zero training error) must have misclassification error $< \epsilon$.

Probabilistic Bounds

Now let S be the training set, we will draw two samples from S as follows:

- First draw a sample S_1 of size m, let A be the event that S_1 fails to be an ϵ-net for $\Delta_f(\mathcal{C})$.
- This means $\exists \tilde{f} \in \Delta_{f,\epsilon}(\mathcal{C})$ such that $\tilde{f}(\boldsymbol{x}) = 0$ for all $\boldsymbol{x} \in S_1$, but has misclassification rate at least ϵ.
- Fix such a $\tilde{f}$ and draw a second sample S_2 of size m ($S_1 \cap S_2 = \varnothing$), then we know $\mathbb{P}(\tilde{f}(\boldsymbol{x}_i) = 1) \geq \epsilon$ for all $\boldsymbol{x}_i \in S_2$, and we have

$$X = \sum_{i=1}^{m} \tilde{f}(\boldsymbol{x}_i) \sim Binomial(m, p), p \geq \epsilon$$

we are interested in the probability

$$\mathbb{P}\left(X \leq \frac{\epsilon m}{2}\right)$$

Probabilistic Bounds

Lemma (Lower Tail Chernoff Bound)

Let $X \sim \textit{Binomial}(m, p)$, $\mathbb{E}X = \mu$, *for all* $0 < \delta < 1$,

$$\mathbb{P}(X \leq (1-\delta)\mu)) \leq \exp\left(-\frac{\delta^2 \mu}{2}\right)$$

Thus by this lemma, we can show that

$$\mathbb{P}\left(X \leq \frac{\epsilon m}{2}\right) \leq \frac{1}{2},$$

whenever $m \geq (8 \ln 2)/\epsilon$.

Probabilistic Bounds

- Now consider the event B as follows: A sample $S = S_1 \cup S_2$ of size $2m$ with $|S_1| = |S_2| = m$, with the classifier $\tilde{f}$ we fixed such that $|\{\boldsymbol{x} \in S, \tilde{f}(\boldsymbol{x}) = 1\}| \geq \epsilon m/2$ and $\tilde{f}(\boldsymbol{x}) = 0$ for all $\boldsymbol{x} \in S_1$.
- We have already showed that,

$$\mathbb{P}(B|A) = \mathbb{P}(X \geq \epsilon m/2) \geq \frac{1}{2}$$

hence

$$\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A) \geq \frac{1}{2}\mathbb{P}(A)$$ **(Vapnik and Chervonenkis, 1971)**

Probabilistic Bounds

Now it is purely a combinatorial problem to bound $\mathbb{P}(B)$. B is equivalent as saying: If we are given $2m$ balls with $r \geq \epsilon m/2$ red and the rest is black, then we divide them into 2 sets of m balls each, what is the probability that the first set has no red balls? The answer is simply $\binom{m}{r}/\binom{2m}{r}$, which can be bounded by

$$\frac{\binom{m}{r}}{\binom{2m}{r}} := \prod_{i=0}^{r-1} \frac{m-i}{2m-i} \leq \frac{1}{2^r}$$

Thus by Sauer's lemma we finally have

$$\mathbb{P}(A) \leq 2\mathbb{P}(B) \leq 2 \cdot \left|\Pi_{\Delta_{f,\epsilon}(\mathcal{C})}(S)\right| 2^{-\frac{\epsilon m}{2}} \leq 2 \cdot \left(\frac{2em}{d}\right)^d 2^{-\epsilon m/2}$$

Probabilistic Bounds

From the bound we obtained, for some δ, we solve

$$2\left(\frac{2em}{d}\right)^d 2^{-\epsilon m/2} \leq \delta$$

and we get

$$\frac{\epsilon m}{2} \geq \frac{d}{\ln 2}\ln\left(\frac{2em}{d}\right) + 1 - \log_2 \delta$$

By choosing m large enough, for instance

$$m \geq \kappa_0\left(\frac{1}{\epsilon}\log\frac{1}{\delta} + \frac{d}{\epsilon}\log\frac{1}{\epsilon}\right)$$

where κ_0 is some universal constant, we would have $\mathbb{P}(A) \leq \delta$. It means that, if we draw at least m training samples, then with probability at least $1-\delta$, every consistent classifier f on S will have misclassification error $< \epsilon$.

Probabilistic Bounds for Testing Error

Recall that a classifier is a Borel measurable function $f : \mathcal{X} \to \mathcal{Y} := \{0, 1\}$, given training set $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$ we define

$$\hat{f} := \arg\min_{f \in \mathcal{C}} \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{f(\boldsymbol{x}_i) \neq y_i\}$$

i.e., it is the classifier in $\mathcal{C}$ that minimizes the empirical misclassification rate over the training set. With the assumption that $\mathcal{Y} = f(\mathcal{X})$, we actually have $R(\hat{f}) = 0$, that is, the empirical risk minimization would be zero on the training set. We would always achieves zero training error, but then we would like to ask the misclassification error overall. We denote $R(\hat{f})$ as the testing error of $\hat{f}$.

Probabilistic Bounds for Testing Error

Theorem (Vapnik and Chervonenkis, 1971)

Let $\mathcal{C} := \{f : \mathcal{X} \to \{0,1\}\}$ be a class such that $\mathbf{VCD}(\mathcal{C}) = d$. Given a training sample $\mathcal{D}_n = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$, let $\hat{f}$ be an ERM classifier (minimizes the empirical risk), then for all probability distribution P over $\mathcal{X} \times \mathcal{Y}$ that satisfies $\mathcal{Y} = f(\mathcal{X})$ for some $f \in \mathcal{C}$, for any $n \geq d, \epsilon > 0$,

$$\mathbb{P}(R(\hat{f}) \geq \epsilon) \leq 2 \left(\frac{2en}{d}\right)^d e^{-\epsilon n/2}$$

The theorem is equivalent as saying, for any $n \geq d$, with probability at least $1 - \delta$,

$$R(\hat{f}) \leq \frac{2\left(d \log \frac{2en}{d} + \log \frac{2}{\delta}\right)}{n}$$

Uniform Convergence of Empirical Risk

If we define $R(f)$ as the true misclassification rate, and $\hat{R}(f)$ as the empirical misclassification rate It can be shown that

Theorem (Vapnik and Chervonenkis, 1971)

$$\mathbb{P}\left(\sup_{f\in\mathcal{C}}|R(f)-\hat{R}(f)|>\epsilon\right)\leq 8\left(\frac{en}{d}\right)^{d}e^{-n\epsilon^{2}/32}$$

for any distribution P on $\mathcal{X}\times\mathcal{Y}$, any $n\geq d, \epsilon>0$.

Hence we have

$$\sum_{n=1}^{\infty}\mathbb{P}\left(\sup_{f\in\mathcal{C}}|R(f)-\hat{R}_n(f)|>\epsilon\right)\leq\sum_{n=1}^{\infty}Cn^{d}e^{-n\epsilon^{2}/32}<\infty$$

which guarantees the uniform convergence by Borel-Cantelli lemma, hence

$$\sup_{f\in\mathcal{C}}|R(f)-\hat{R}_n(f)|\to 0 \quad a.s$$

References

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- **2.** On the Uniform Convergence of Relative Frequences of Events to Their Probabilities, by V.N. Vapnik and A. Ya. Chervonenkis;
- **3.** Machine Learning Theory (CSC 482A/581B), Lectures 7 & 8, by Nishant Mahta;